

CAPSTONE PROJECT TITLE APPROVAL FORM

Proponents:

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{Role}

Project Manager

Researcher

Document Writer

Developer

Designer

Capstone Project Title: Disaster Response Evacuee and Resource Allocation System (DRERAS) for Dagupan city

Project Type: **{select the nature of project}**

- ☐ System Development
- ☐ Mobile Application
- ☐ Game Development
- ☐ Cybersecurity Framework
- ☐ Network Infrastructure

Evaluation

{For completion by Dean Aris Liwanag}

	Approved	Disapproved
Feasibility		
Relevance to the Program		
Originality and Innovation		
Alignment with SDGs		

Overall Recommendation

- ☐ Approved
- ☐ Disapproved

Instructions for re-submission (if disapproved)

Recommending Approval:

Approved by:

Dean Aristotle B. Liwanag

PROJECT INFORMATION	
RESEARCH TITLE	Disaster Response Evacuee and Resource Allocation System (DRERAS) for Dagupan City
TECHNICAL DESCRIPTION	
RATIONALE	Dagupan City is highly prone to flooding, storm surges, and typhoons, making disaster preparedness and response a critical priority. However, many evacuation centers still rely on manual paper-based recording, leading to delays, inaccurate reporting, and misallocation of relief goods. These challenges can compromise public safety during high-pressure disaster operations. The Disaster Response Evacuee and Resource Allocation System (DRERAS) addresses these issues through AI-assisted data entry, real-time household-level tracking, and automated relief prioritization. AI reduces manual errors, speeds up evacuee check-in, and provides recommended resource allocations based on need and barangay-level damage severity. The addition of blockchain-based record integrity ensures transparent and tamper-proof logs of evacuee registration and relief distribution, increasing trust among responders, barangays, and LGUs. This project supports the following Sustainable Development Goals: ● SDG 11 – Sustainable Cities and Communities (Strengthening disaster resilience) ● SDG 13 – Climate Action (Improving adaptation and response mechanisms) ●
OBJECTIVES	● To develop an AI-assisted evacuee check-in and household monitoring system for use in evacuation centers. ● To design a municipal dashboard that visualizes barangay-level damage severity, evacuee statistics, and resource allocation needs. ● To implement blockchain-based verification for recording check-ins and relief distribution events to ensure data integrity. ● To integrate offline-first functionality that allows core operations to continue with or without internet connectivity. ● To evaluate the system's accuracy, usability, and efficiency through user testing with MDRRMO staff and barangay responders.
METHODOLOGY	1. Data Gathering Conduct interviews and surveys with barangay officials, MDRRMO staff, and evacuation center personnel to understand existing workflows. 2. System Design Create UI/UX wireframes, database architecture, and feature workflows.

	3. Development - Implement household module, evacuee check-in system, AI-based suggestions, dashboard, and offline capability. - Develop blockchain smart contract or hashing mechanism to ensure tamper-proof logs of key system events. 4. Testing and Evaluation Perform system testing, debugging, user acceptance testing, and optimization.
REFERENCES	[1] National Disaster Risk Reduction and Management Council. (2023). Philippine disaster response operations report. https://ndrrmc.gov.ph .
	[2] Cruz, G. T., & Du, M. (2020). Digital transformation in disaster management: Opportunities for local government units. <i>Philippine Journal of Public Administration</i> , 64(2).
	[3] United Nations. (2015). <i>Sustainable Development Goals</i> . https://sdgs.un.org/goals
EXPECTED OUTPUT	- A fully functioning Disaster Response Evacuee and Resource Allocation System (DRERAS) - Admin dashboard for municipal disaster managers. - AI-assisted modules for suggestions, completeness checking, and relief prioritization. - Project documentation (introduction, system design, methodology, evaluation). - User manual for staff and responders. - Blockchain-based evacuee and relief distribution logging system. - Testing and evaluation results.
POTENTIAL IMPACT	- Faster and more accurate evacuee check-ins during chaotic disaster situations. - Improved allocation of food, water, and resources through AI prioritization. - Better visibility of barangay-level impact, enabling quicker decision-making. - Greater transparency and prevention of duplicate relief claims using blockchain-based logs.
USERS OR BENEFICIARIES	- Municipal Disaster Risk Reduction and Management Offices (MDRRMO). - Barangay officials and evacuation center personnel. - Evacuees and affected households. - Local government units managing relief distribution.

